Project 2 Written Portion

Wednesday, March 25, 2026 2:36 PM

1 Solve Quadratic Equation

Written Part: Write the **algorithm or thinking process** you will follow to design your code for solving a quadratic equation. Explain step by step how you will:

- compute the discriminant $\Delta = b^2 - 4ac$,
- check whether Δ is positive, zero, or negative,
- decide whether the roots are real and distinct, real and equal, or imaginary.

For this problem, I need to create a function called "quadroots" to call upon in main. Should take 3 inputs, a, b, & c.

Pseudocode for quadroots:

```
function quadroots(a, b, c){
  % Compute discriminant
    discriminant = b^2 - 4*a*c
  % Is the discriminant +, -, or 0?
  If discriminant > 0
     Then print "Discriminant is positive."
  Else if discriminant < 0
     Then print "Discriminant is negative."
  Else if discrimnant = 0
     Then print "Discriminant is zero."
  % Determine the roots of the function, if any.
    x1 = (-b + √discriminant)/(2*a);
    x2 = (-b - √discriminant) / (2*a);

  If x1 = error
     Then print "No root for x1."
  Else print x1
  If x2 = error
     Then print "No root for x2."
```

If x_2 = error
Then print "No root for x_2."
Else print x_2

% Determine if roots are real and equal, real and distinct, or imaginary
If $x_1 = x_2$
Then print "The roots are real and equal."
Else If $x_1 \neq x_2$
Then print "The roots are real but distinct."
Else If x_1 & x_2 are imaginary
Then print "The roots are imaginary."
end
} % end of function

2 Count Positive, Negative, and Zero Elements

Written Part: Write the **algorithm or step-by-step logic** you will use in your function to count positive, negative, and zero elements in a matrix. Briefly explain how you will use `if`, `elseif`, and `else` inside loops to classify each entry.

Start code with the function "countSigns(A)". The function should take numeric values for matrix A & return the following:

- # of + elements
- # of − elements
- # of 0 elements

Pseudocode for countSigns:

```
function [n_pos, n_neg, n_0] = countSigns(A) {
  % Counts for all 3 variables must be assigned to 0
  n_pos = 0;
  n_neg = 0;
  n_0 = 0;

  % Use nested for loops to get # of rows & columns of matrix A

  for i = 1 : size(A,1)
```

matrix A

```
for i = 1 : size(A,1)
    for j = 1 : size(A,2)
        % Use an if loop inside to determine if the element
          is positive, negative, or 0 and add to count
        if A(i,j) > 0
            n_pos = n_pos + 1
        else if A(i,j) < 0
            n_neg = n_neg + 1
        else
            n_0 = n_0 + 1
        end
    end
}end of function
```

3 Linear Transformations in $\mathbb{R}^2$

Written Part: Using the **standard basis vectors**

$$e_1 = (1,0), \qquad e_2 = (0,1),$$

build the **standard matrix** for each of the following transformations on $\mathbb{R}^2$ by finding the images of e_1 and e_2.

1. Rotation by an angle θ (counterclockwise)
2. Reflection across the x-axis
3. Stretch along the x-axis by a given value (keeping y the same)
4. Shear along the x-axis by a given value
5. Projection onto the y-axis

The standard matrix $A = [T(e_1) \quad T(e_2)]$

Rotation by θ CCW:

$$A = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

Reflect across x-axis:

$$\left.\begin{aligned} T(e_1) &= \begin{bmatrix} 1 \\ 0 \end{bmatrix} \\ T(e_2) &= \begin{bmatrix} 0 \\ 1 \end{bmatrix} \end{aligned}\right\} \Rightarrow A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\mathcal{T}(e_2) = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \Big)$$

Stretch along x-axis (by a):

$$\left.\begin{aligned} \mathcal{T}(e_1) &= \begin{bmatrix} a \\ 0 \end{bmatrix} \\ \mathcal{T}(e_2) &= \begin{bmatrix} 0 \\ 1 \end{bmatrix} \end{aligned}\right\} \Rightarrow A = \begin{bmatrix} a & 0 \\ 0 & 1 \end{bmatrix}$$

Shear along x-axis (by k):

$$\left.\begin{aligned} \mathcal{T}(e_1) &= \begin{bmatrix} 1 \\ 0 \end{bmatrix} \\ \mathcal{T}(e_2) &= \begin{bmatrix} k \\ 1 \end{bmatrix} \end{aligned}\right\} \Rightarrow A = \begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix}$$

Projection onto y-axis:

$$\left.\begin{aligned} \mathcal{T}(e_1) &= \begin{bmatrix} 0 \\ 0 \end{bmatrix} \\ \mathcal{T}(e_2) &= \begin{bmatrix} 0 \\ 1 \end{bmatrix} \end{aligned}\right\} \Rightarrow A = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

4 Column Space and Null Space

Written Part:

Consider the matrix

$$A = \begin{bmatrix} 1 & -2 & 0 & 4 \\ 2 & -4 & 1 & 9 \\ -1 & 2 & 1 & -1 \end{bmatrix}.$$

Using the RREF computed in MATLAB (see the MATLAB part below), fill in the following table in your written report:

Quantity	**Your Answer**
rank(A)	3
nullity(A)	1
Verify: rank(A) + nullity(A) = n	$3 + 1 = 4$
Basis for Col(A)	(write each basis vector) $\begin{bmatrix} 1 & 0 & 4 \\ 2 & 1 & 9 \\ -1 & 1 & -1 \end{bmatrix}$
Basis for Nul(A)	(write each basis vector) $\begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \end{bmatrix}$